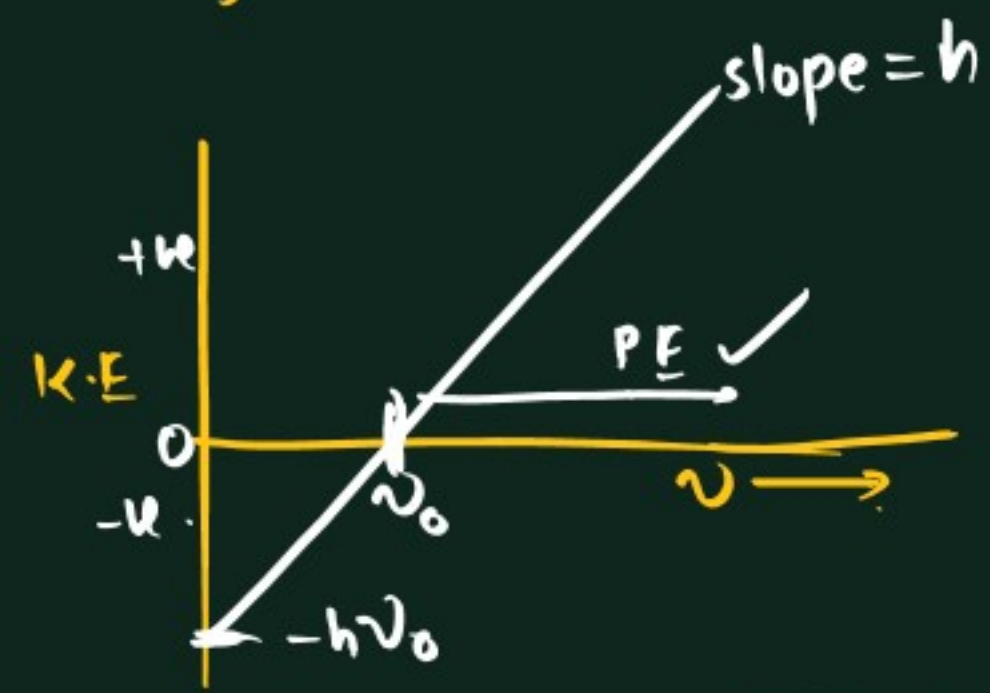


Graph $\rightarrow$

a) $K.E_{max}$ vs ν
 y vs x



$$K.E_{max} = h\nu - h\nu_0$$

y m $x +$ c
 $= +ve$

$$y = mx + c$$

slope intercept on y-axis

+ve -ve

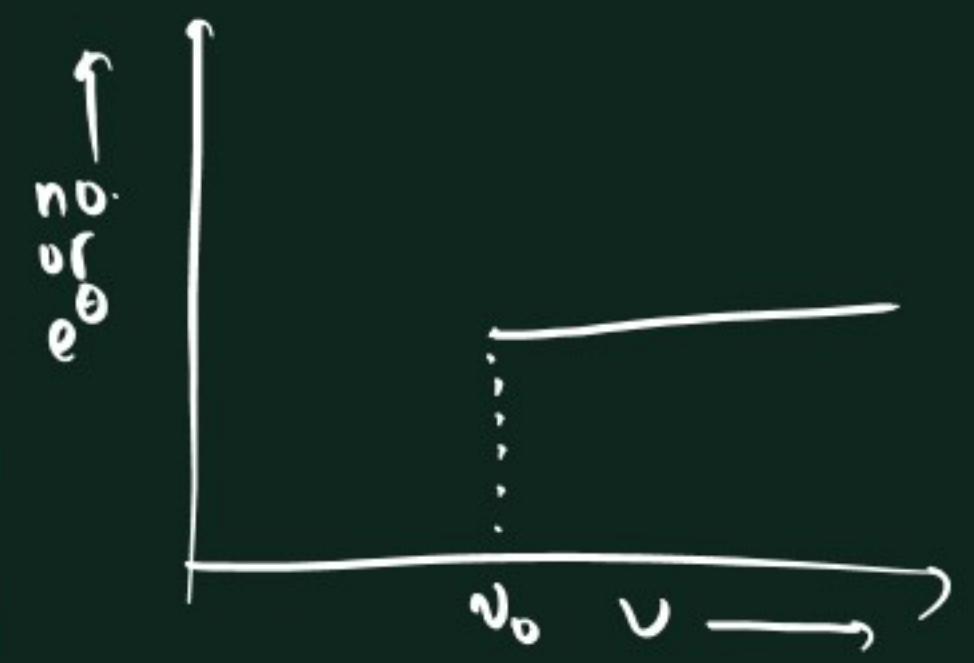
$\nu \uparrow \Rightarrow K.E \uparrow$

$$E = E_0$$

$$h\nu = h\nu_0$$

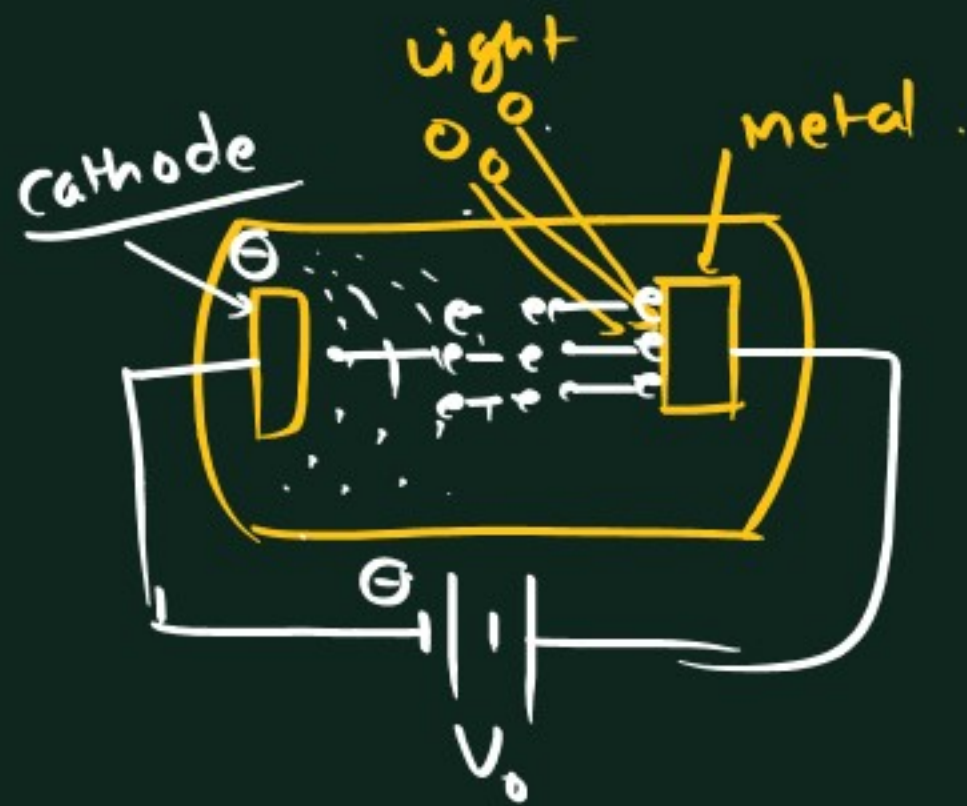
$$\nu = \nu_0$$

2) no. of photo e^- vs ν
 As $\nu \uparrow \Rightarrow$ no. of photo $e^- \rightarrow$ const.



3) If Intensity of light $\uparrow \Rightarrow$ no. of photoe^s $\uparrow$
(no. of photons)

* stopping potential (V_0):



$K.E = eV_0$

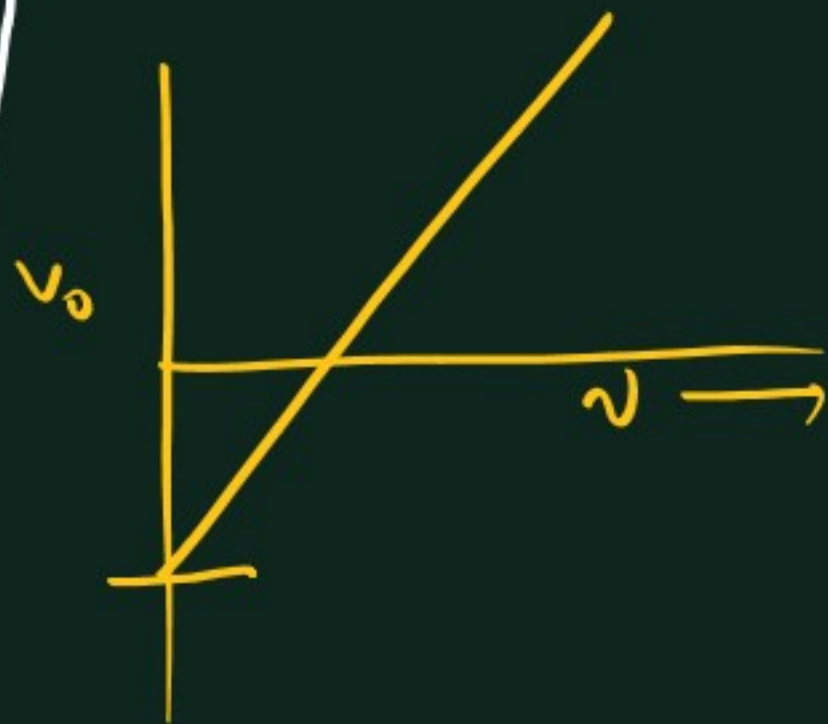
charge of e

stopping potential.

IF $v \uparrow \Rightarrow K.E_{max} \uparrow \Rightarrow v_0 \uparrow$

$$v_0 \quad v_s \quad v$$
$$K.E = h\nu - h\nu_0$$
$$eV_0 = h\nu - h\nu_0$$

$$V_0 = \frac{h\nu}{e} - \frac{h\nu_0}{e}$$



* Bohr's theory →

→ only applicable to 1 e^- system

e.g. H, D, T, He^+ , Li^{+2} , Be^{+3} , O^{+7} , No^{+10} etc

①



shell or orbit (n) or energy level

n	notation	$\text{max}^m e^- = 2n^2$
1	K	2
2	L	8
3	M	18
4	N	32

②

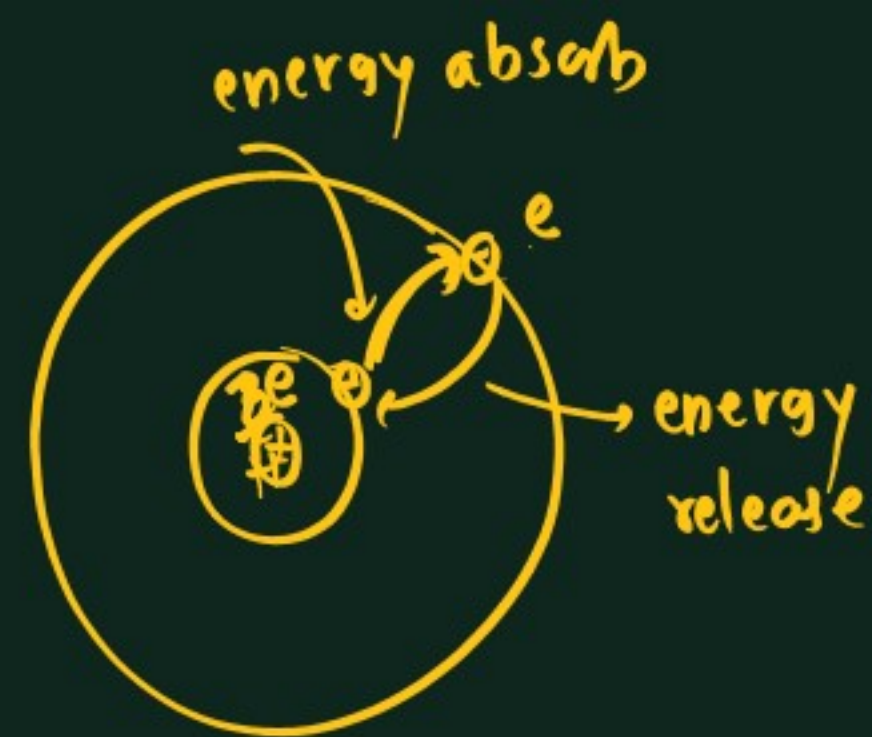
③ $E_3 > E_2 > E_1$

$$k = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

$q_1 = \text{charge of } e^- = e$

$q_2 = \text{charge of Nu} = Ze$

④



⑤



Electrostatic FOA = Centrifugal Force

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$\frac{kZe^2}{r^2} = \frac{mv^2}{r}$$

$e \rightarrow$ charge of $e^\ominus$

$m \rightarrow$ mass of $e^\ominus$

$z \rightarrow$ atomic no.

$r \rightarrow$ radius of shell/orbit.

$v \rightarrow$ velocity of $e^\ominus$

(c) Angular momentum of $e^\ominus$ is quantized in nature.

Angular momentum of an $e^\ominus = \frac{nh}{2\pi}$

(mvr)

$$mvr = \frac{nh}{2\pi}$$

$n \rightarrow$ orbit no.

n	$mvr = \frac{nh}{2\pi}$
1	$\frac{h}{2\pi}$
2	$\frac{2h}{2\pi}$
3	$\frac{3h}{2\pi}$

$$* \quad mvr = \frac{2h}{3\pi}$$

$$mvr = \frac{nh}{2\pi}$$

$$\frac{2h}{3\pi} = \frac{nh}{2\pi}$$

$$n = \frac{4}{3}$$

$$\frac{kze^2}{r^2} = \frac{mv^2}{r} \quad \text{--- (1)}$$

$$mvr = \frac{nh}{2\pi} \quad \text{--- (2)}$$

$$v = \frac{nh}{2\pi mr}$$

$$r = \frac{n^2 h^2}{4\pi^2 m k z e^2}$$

$$r = \left(\frac{h^2}{4\pi^2 m k e^2} \right) \frac{n^2}{z}$$

$$r = 0.529 \frac{n^2}{z} \text{ \AA}$$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

4 Eqⁿ of velocity -

$$V = \frac{kze^2 2\pi}{nh}$$

$$V = \frac{ke^2 2\pi}{h} \left(\frac{z}{n} \right)$$

$$V = 2.18 \times 10^8 \times \frac{z}{n} \frac{\text{cm}}{\text{sec}}$$

$$r_1 = 0.52g \frac{n_1^2}{z_1}$$

$$r_2 = 0.52g \frac{n_2^2}{z_2}$$

$$\frac{r_1}{r_2} = \frac{\frac{n_1^2}{z_1}}{\frac{n_2^2}{z_2}} = 3/8$$

$$= \frac{(1)^2/2}{(2)^2/3} = \frac{1/2}{4/3} = 3/8$$

$$\begin{aligned} \textcircled{3} \quad \frac{r_2 - r_1}{r_5 - r_4} &= \frac{\frac{0.52g \times n_2^2}{2} - \frac{0.52g \times n_1^2}{2}}{\frac{0.52g \times n_5^2}{2} - \frac{0.52g \times n_4^2}{2}} \\ &= \frac{4-1}{25-16} = \frac{3}{9} = 1/3 \end{aligned}$$

$$\textcircled{4} \quad \textcircled{a_0} \quad a_0 = 0.52g \frac{(1)}{1}$$

$$\begin{aligned} r &= \frac{0.52g (3)^2}{4} \\ &= \underbrace{0.52g \times 9}_{\frac{9}{4} a_0} \\ &= \frac{9}{4} a_0 \end{aligned}$$

$$4.76 \text{ Å}^0$$

$$r = 0.529 \frac{n^2}{Z}$$

$$4.76 = 0.529 \times n^2$$

$$n^2 = \frac{4.76}{0.529} \approx 9$$

$$n^2 = 9$$

$$n = 3$$

* Time required for 1 revolution —

(T)

$$v = \frac{\text{distance}}{\text{time}}$$

$$\text{time} = \frac{\text{distance}}{\text{velocity}}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi \left(\frac{1}{2} \frac{n^2}{Z} \right)}{\left(\frac{1}{2} \frac{Z}{n} \right)} = 1.52 \times 10^{-16} \frac{n^3}{Z^2} \text{ sec}$$

$$\begin{aligned} v &= 2.18 \times 10^8 \frac{Z}{n} \\ &= 2.18 \times 10^8 \times \frac{1}{3} \\ &= 0.72 \times 10^8 \frac{\text{cm}}{\text{sec}} \end{aligned}$$

freqⁿ of revolution
(f)

$$f = \frac{1}{T} = \frac{1}{1.52 \times 10^{-16} \times \frac{n^3}{Z^2}}$$

$$= 0.65 \times 10^{16} \frac{Z^2}{n^3} \frac{1}{\text{sec}} \text{ or Hz}$$

Energy of $e^- \rightarrow$

$$E = \underbrace{P.E}_{\text{Total energy}} + \underbrace{K.E}_{\text{kinetic energy}} \quad \text{potential energy}$$

* $P.E \rightarrow$ force $\times$ distance.

$$= -\frac{kze^2}{r^2} \times r$$

$$= -\frac{kze^2}{r}$$

* $K.E \rightarrow$

$$\frac{kze^2}{r^2} = \frac{mv^2}{r}$$

$$\frac{1}{2} \frac{kze^2}{r} = \frac{1}{2} mv^2$$

$$K.E = \frac{kze^2}{2r}$$

$$\begin{aligned} E &= P.E + K.E \\ &= -\frac{kze^2}{r} + \frac{kze^2}{2r} \\ &= \frac{kze^2}{r} \left(-1 + \frac{1}{2}\right) \end{aligned}$$

$$E = -\frac{kze^2}{2r}$$

$$E = -K.E$$

$$E = \frac{P.E}{2}$$

$$P.E = 2E$$

$$P.E = -2K.E$$

$$E = -\frac{kze^2}{2r}$$

$$E = -\frac{2mk^2z^2e^4\pi^2}{n^2h^2}$$

$$= -\frac{2mk^2e^4\pi^2}{h^2} \left(\frac{z^2}{n^2}\right)$$

$$E = -13.6 \times \frac{z^2}{n^2} \frac{\text{eV}}{\text{atom}}$$

$$E = -2.18 \times 10^{-18} \times \frac{z^2}{n^2} \text{ J/atom}$$

$$E = -313 \frac{z^2}{n^2} \frac{\text{kcal}}{\text{mol}}$$

for H-atom $E = -\frac{13.6 \times 1}{n^2}$

n	$E(\text{eV})$
1	-13.6
2	-3.4
3	-1.5
4	-0.85
5	-0.54
...	...
∞	0

energy of e^- increases

max^m energy of $e^- = 0$ at $n \rightarrow \infty$ that shows e^- escapes from that atom.

1) $E = -6.04 \text{ eV}$

2) $E_1 = -\frac{13.6 \times Z_1^2}{n_1^2}$

$E_2 = -\frac{13.6 \times Z_2^2}{n_2^2}$

$\frac{E_1}{E_2} = \frac{Z_1^2/n_1^2}{Z_2^2/n_2^2} = \frac{4}{81}$

3) $E = -\frac{13.6 \times Z^2}{n^2} = -\frac{13.6 \times (4)^2}{(2)^2} = -108.8 \text{ eV}$

④

$$E_0 = \frac{-13.6 \times 1}{9}$$

$$-13.6 = 9E_0$$

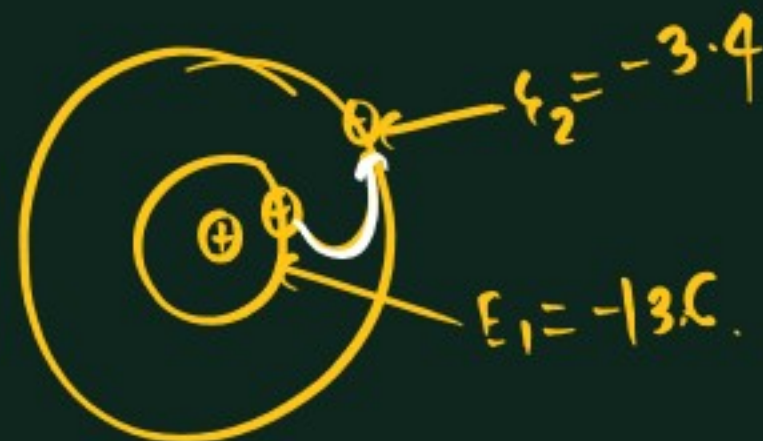
$$E = \frac{-13.6 \times 9}{4}$$

$$= \frac{9E_0 \times 9}{4}$$

$$= \frac{81E_0}{4}$$

* Energy of photon absorb or release during transition of e^- from b/w two shells.

→ n_1 → lower shell
 n_2 → higher shell



$$\Delta E = E_2 - E_1$$

energy of photon

$$= \frac{-13.6 \times Z^2}{n_2^2} - \left(\frac{-13.6 \times Z^2}{n_1^2} \right)$$

$$= \frac{-13.6 \times Z^2}{n_2^2} + \frac{13.6 \times Z^2}{n_1^2}$$

$$\Delta E = +13.6 \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

$$\Delta E = 2.18 \times 10^{-18} \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J}$$

$n=1$ → ground state

$n=2$ → 1st excited state

$n=3$ → 2nd excited state

excited state → $n-1$

n = excited state + 1

- 1) when e^- goes from ground state $\rightarrow$ 1st excited state $\rightarrow$ energy absorb is known as 1st excitation potential
- 2) -11 $\xrightarrow{n=3}$ 2nd excited state $\rightarrow$ energy -11 $\xrightarrow{-11}$ 2nd excitation potential
- 3) when e^- goes from ground state to $\infty \rightarrow$ energy $+1$ $\xrightarrow{-11}$ Ionisation potential
- 4) when e^- -11 any excited state to $\infty \rightarrow$ energy -11 $\xrightarrow{-11}$ separation energy

Int. Ex-03

In Ex-01 - 13-22

Ex-02 - 11-21